

SURNAME NAME MAT. NUMBER

Signature _____

1

Consider the following DAE system.

$$\begin{cases} x'(t) = y(t) + t^2 \\ x(t) = t \\ x(t)z(t) = 1 \end{cases}$$

rewrite in the form

$$\mathbf{E}(\mathbf{x}, t)\mathbf{x}' = \mathbf{G}(\mathbf{x}, t)$$

perform index reduction

$$\mathbf{E}^{(1)}(\mathbf{x}, t)\mathbf{x}' = \mathbf{G}^{(1)}(\mathbf{x}, t), \quad \dots \quad \mathbf{E}^{(k)}(\mathbf{x}, t)\mathbf{x}' = \mathbf{G}^{(k)}(\mathbf{x}, t),$$

until $\mathbf{E}^{(k)}(\mathbf{x}, t)$ is non singular and DAE is reduced to an ODE. Compute:

Differential index:

List of all invariants (or algebraic constraints):

Matrices $\boldsymbol{E}^{(j)}(\boldsymbol{x}, t)$ and vector $\boldsymbol{G}^{(j)}(\boldsymbol{x}, t)$ at each step

2

Consider the following DAE system:

$$\begin{cases} x''(t) = \lambda(t)y(t) \\ y''(t) = \lambda(t)x(t) + t \\ x(t)y(t) = 1 + t \end{cases}$$

reduce to order 1 then using dummy variable reduce the DAE to index 1. Compute:

Differential index:

DAE rewritten using dummy variables

List of all invariants (or algebraic constraints):

DAE reduced to index 1

Projection step to $xy = 1 + t$ (first invariant)